

{HW3 Short}

1. Define $\mathcal{H} = \{x \mapsto \text{sign}(w^t x) : w \in \mathbb{R}^d\}$, the hypothesis class of homogeneous linear classifiers.

1.1. In Tutorial 05, we said that the VC-dimension of homogeneous linear classifiers is $\geq d$.

Provide a rigorous proof for this statement.

1.2. Prove that $VCdim(\mathcal{H})$ is exactly d by proving that $VCdim(\mathcal{H}) < d + 1$

Answer:

1.1 To prove that $VCdim(\mathcal{H}) \geq d$, we need to show that there exists a set C of size d that $\mathcal{H}$ shatters.

Shattering means that for every possible labelling y_i of these d points, there exists a classifier in $\mathcal{H}$ that correctly separates the points according to that labeling.

And indeed, denote C the set consisting of the standard basis vectors $e_i \forall i \in [d]$.

Let a labelling $(y_1, \dots, y_d) \in \{-1, +1\}^d$, since each classifier $h_w \in \mathcal{H}$ is defined by $w \in \mathbb{R}^d$, taking the classifier defined by

$w = (y_1, \dots, y_d)$ results in the following:

$\forall e_i \in C, h_w(e_i) = \text{sign}(w^t e_i) = \text{sign}(y_i) = y_i$.



1.2 Assume by contradiction $\mathcal{H}$ shatters any set of $d+1$ points denoted $x_i \in \mathbb{R}^d, i \in [d+1]$. Since x_i are linearly dependent,

$$\exists a_i \in \mathbb{R}, \text{ not all zero such that: } \sum_{i=1}^{d+1} a_i x_i = 0.$$

Without loss of generality, assume $\sum_{i=1}^d a_i x_i = x_{d+1}$ for some $a_i \in \mathbb{R}$.

Since $\mathcal{H}$ shatters all Consider $w \in \mathbb{R}^d$ **Fitting for** $C = \{x_{d+1}\} \cup \{a_i x_i\}_{i=1}^d$ with the labels $y_{d+1} = -1, y_i = +1 \forall i \in [d]$.

And now, since $\sum_{i=1}^d a_i x_i = x_{d+1}$, then $w^t \sum_{i=1}^d a_i x_i = w^t x_{d+1}$

By the linearity of dot product: $\sum_{i=1}^d w^t (a_i x_i) = w^t x_{d+1}$.

Now, taking the sign of both sides we get that the sign of the right hand side is -1 since $y_{d+1} = -1$, and the sign of the left hand side is -1, since the sign of each element in the sum is +1 since $y_i = +1, \forall i \in [d]$. Leading to a contradiction.

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2. Let $\phi: \mathcal{X} \rightarrow \mathbb{R}^{n_1}, \phi': \mathcal{X} \rightarrow \mathbb{R}^{n_2}$ be two feature mappings where $n_1, n_2 \in \mathbb{N}$.

Let $K, K': (\mathcal{X} \times \mathcal{X}) \rightarrow \mathbb{R}$ be two **valid kernels** defined as:

$$K(u, v) = \langle \phi(u), \phi(v) \rangle, \quad K'(u, v) = \langle \phi'(u), \phi'(v) \rangle$$

Proving that $G(u, v) \triangleq K(u, v) \cdot K'(u, v)$ is a valid kernel.

Answer: let $n_3 = n_1 * n_2$, define $\psi: \mathcal{X} \rightarrow \mathbb{R}^{n_3}$ by:

$$\psi(u) = \phi(u) * \phi'(u)^t$$

i.e. the outer product of $\phi(u)$ and $\phi'(u)$.

Now define:

$\psi(u) = \text{flatten_rows}(\psi(u))$ to be from $\mathcal{X}$ to an n_3 long column vector

Observe:

$$\begin{aligned} \langle \psi(u), \psi(v) \rangle &= \psi(u)^t \psi(v) = \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} (\psi(u))_{j,i} (\psi(v))_{i,j} = \\ &= \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} (\phi_j(u) \phi'_i(u)) (\phi_i(v) \phi'_j(v)) = \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \phi_j(u) \phi_i(v) \phi'_i(u) \phi'_j(v) \\ &= \left(\sum_{j=1}^{n_1} \phi_j(u) \phi_i(v) \right) \left(\sum_{j=1}^{n_2} \phi_j(u) \phi'_j(v) \right) = K(u, v) * K'(u, v) \end{aligned}$$

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3. For a given parameter $\gamma > 0$, define the Gaussian Kernel for 1-D input in the following manner:

$$K: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}, K(a, b) = \exp(-\gamma(a - b)^2)$$

3.1. Provide a feature mapping $\phi: \mathbb{R} \rightarrow \mathbb{R}^p$, with $p \in \mathbb{N} \cup \{\infty\}$, and prove that K is indeed a valid kernel.

3.2. Assume that you are given a very large dataset with 1-D samples. We would like to apply the

Gaussian Kernel to train a classifier on the dataset. Would it be better to optimize the **primal problem** with the feature mapping you found, or is it better to optimize the **dual problem** with the kernel that we defined? Is it even possible? Explain.

Answer:

3.1 define $\phi: \mathbb{R} \rightarrow \mathbb{R}^p$ for $p = \text{infinity}$ such that:

$$\phi(x) = e^{-\gamma x^2} \left(1, \sqrt{\frac{(2\gamma)}{1!}} a, \dots, \sqrt{\frac{(2\gamma)^i}{i!}} a^i, \dots, \dots \right), \text{ Now, consider the product}$$

$$\phi(x)^t \phi(y) = e^{-\gamma(x^2+y^2)} \sum_{i=1}^{\infty} \frac{(2\gamma xy)^i}{i!} = e^{-\gamma(x^2+y^2)+2\gamma(xy)} = e^{-\gamma(x-y)^2}$$

$$\text{We get that } K(a, b) = \phi(a)^t \phi(b), \quad \forall (a, b) \in \mathbb{R}^2$$

therefore a valid kernel.



3.2 we choose the dual problem with the defined kernel because of the kernel trick, providing benefits for Efficiency and feasibility.

- Efficiency:

For very large datasets, optimizing the dual problem is often more efficient because it avoids the actual computation and storage of the high-dimensional feature vectors.

Instead, we use the kernel matrix directly, which is more practical for large-scale problems.

- Feasibility:

It is possible to optimize the dual problem with the Gaussian Kernel. The kernel trick makes it computationally feasible to handle the infinite-

dimensional feature space. Which is otherwise memory impossible optimizing the primal problem.

4. Refute (with a simple example): Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be two convex functions.

The composition $h \triangleq f \circ g$ is also a convex function.

Answer: consider the following example:

$$f(x) = -x + 4, g(x) = x^2$$

f is convex since it is linear, g is convex since $g''(x) = 2 > 0$.

Yet $h(x) = -x^2 + 4$ is not convex, it is even concave.

5. Let $f, g: C \rightarrow \mathbb{R}$ be two convex functions defined over a convex set C .

Lemma (no need to prove): $q(z) \triangleq \max\{f(z), g(z)\}$ is convex w.r.t z .

Lemma (no need to prove): the sum of any number of convex functions is convex.

5.1. Prove (by definition): Given a constant $\alpha \geq 0$, the function $\alpha f(z)$ is convex w.r.t z .

5.2. Using a rule from Tutorial 07, conclude that $\max\{0, 1 - y_i w^t x_i\}$ is convex w.r.t w .

5.3. Using the above (and properties from Tutorial 07), conclude that the Soft-SVM optimization

problem is convex w.r.t w .

5.1

Let $x_1, x_2 \in \mathbb{R}$, and let $t \in [0, 1]$ consider:

$$\alpha f(tx_1 + (1-t)x_2) \leq \alpha (tf(x_1) + (1-t)f(x_2)) =$$

$$\alpha tf(x_1) + \alpha(1-t)f(x_2) = t(\alpha f(x_1)) + (1-t)(\alpha f(x_2))$$



5.2

We know that $h(x) = w^t x$ is linear therefore convex. $\forall y_i \in \{-1, +1\}$ the function $x \rightarrow (1 - y_i w^t x_i)$ is linear therefore also convex.

Since the function that is constant **zero** is also linear we get that taking the max over both functions is also convex by the given lemma.



5.3

First and foremost, the domain $\mathcal{C} = \mathbb{R}^d$ of w is convex. Now all that remains is to show the convexity of the objective function.

Since $\max\{0, 1 - y_i w^t x_i\}$ is convex $\forall i \in [m]$. The sum of them over all i is convex too.

For now we have that $\sum_{i=1}^m \max\{0, 1 - y_i w^t x_i\} = (**)$ is also convex.

For fixed $\alpha = \frac{1}{m}$, α times $(**)$ is also convex. Which is the objective itself.

All in all we have that SoftSVM is a convex optimization problem.

